COLUMN DESIGN CALCULATIONS

Project Name	:	G+1 RB COLUMN DESIGN
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Column-2.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45

Definitions Of Terms:

All forces in units kN and m

All reinforcement details like area, spacing are mm

Grade Of steel for 6 mm dia. bars is Fe250. It is irrespective of the grade of steel defined by user in column design.

Neutral axis angle for resultant design moment is with respect to local major axis.

1	β	=	Stiffness proportion factor at joint (column stiffness/sum of stiffness of all members connected at joint)
2	Δu	=	Elastically computed first order lateral deflection. (Relative deflections)
3	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
4	εm	=	Average steel strain at level considered
5	acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
6	Ag	=	Gross area of the column cross section
7	Aeff	=	Effective area of column / wall considered for shear check in sqmm
8	Ak	=	Area of confined concrete core
9	Ash	=	Area of link cross section
10	b	=	Effective Width of Column in mm
11	beff	=	Effective Width of Column in mm
12	B	=	Width / Smaller Dimension of Column in mm
13	d	=	Effective Depth of Column in mm
14	deff	=	Effective Depth of Column in mm
15	D	=	Depth / Larger Dimension of Column OR Diameter of Circular Column in mm
16	Dk	=	Diameter Of core measured to the outside of circular link
17	Ec	=	Modulus of elasticity of concrete
18	Es	=	Modulus of elasticity of steel
19	FcPerm	=	Permissible Stress in Concrete required in N/ (sqmm)
20	Fst	=	Stress in steel
21	FstPerm	=	Permissible Stress in Steel required in N/ (sqmm)
22	Fck	=	Characteristic compressive strength of concrete cube

			in N/sqmm
23	Fy	=	Yield Stress Of Steel in N/sqmm
24	h	=	Longer dimension of rectangular link measured to its outer face
25	Hu	=	Total lateral force acting within the story(Story shear)
26	Hs	=	Height of the story (Floor height)
27	k	=	Reduction factor for slenderness moments
28	Max	=	Additional moment due to slenderness about major axis (Along D)
29	May	=	Additional moment due to slenderness about minor axis (Along B)
30	Mminx	=	Moment due to minimum eccentricity along D
31	Mminy	=	Moment due to minimum eccentricity along B
32	Muv	=	Moment capacity of web portion of wall as per clause 9.4.2 and Annes A of IS 13920 - 2016
33	Mux	=	Factored moment Along D (Momemt About Major Axis)
34	Muy	=	Factored moment Along B (Momemt About Minor Axis)
35	MCap	=	Moment capacity of section for NA angle at design Pu
36	MRes	=	Resultant design moment at angle to local major axis
37	Pb	=	Axial capacity of column along B as defined in 39.7.1.1
38	Pd	=	Axial capacity of column along D as defined in 39.7.1.1
39	Pu	=	Factored axial force
40	Puz	=	Axial load on column as defined in 39.6
41	Pu_Total	=	Sum of Axial loads on all column in the story (Gravity Load)
42	Q	=	Stability Index (factor for checking sway/ non sway condition for a given story)
43	S	=	Link Spacing
44	Shear Strength Enhancement Factor	=	Multiplying factor for shear strength of concrete as per 40.2.2
45	Vur	=	Factored resultant shear force acting on the column
46	Vux	=	Factored shear force Along B
47	Vuy	=	Factored shear force Along D
48	Wcr	=	Surface Crack Width
49	WcrPerm	=	Permissible Crack Width required in mm
50	β	=	It is a Neutral Axis angle corresponding to load angle to find out MCap

Code References:

IS 456

	ELEMENT		CLAUSE / table
1	Max area of reinforcement	:	26.5.3.1-a & b
2	Min area of reinforcement	:	26.5.3.1-a & b
3	Min number of bars	:	26.5.3.1-c
4	Minimum Eccentricity Calc	:	25.4 & 39.2
5	Effective Length	:	25.2
6	Slenderness Moments	:	25.3 & 39.7
7	Design for axial loads	:	39.3
8	Design for axial loads And uniaxial bending	:	39.5
9	Design for axial loads And Biaxial bending	:	39.6
10	Design of horizontal links	:	26.5.3.2
11	Design shear strength	:	40.2
12	Stiffness Proportion Factor, β	:	E-1
13	Stability(Index, Q)	:	E -2
14	Crack width calculation	:	3.8
15	Multiplying factor to Beam Stiffness for effective length calculation	:	SP 24 1983 APPENDIX D

IS 13920 - 2016

	ELEMENT		CLAUSE / table
1	Spacing of special confining reinforcement	:	10.4 and 7.6
2	C/s area of special confining reinforcement	:	7.6.1
3	Applicability of boundary element	:	10.4.1
4	Muv-Moment Capacity of Web	:	Annex
5	Additional Compressive Force in BE	:	10.4.2
6	Check for BE in tension and compression	:	10.4.2.1 and 10.4.3
7	Shear include due to Beam	:	7.5
8	Minimum Flexural Strength of Column	:	7.2
9	Shear Check at Column Joint	:	9.1
10	Length of wall to thickness ratio	:	10.1.3
11	Type of wall & Minimum reinforcement	:	10.1.4
12	Largest diameter of longitudinal steel bar	:	10.1.8
13	Shear reinforcement in RC wall	:	10.2.3
14	Special confinement reinforcement in Boundary Element	:	10.4.4
15	Minimum vertical reinforcement across horizontal construction joint	:	10.7

Sway Calculation (Stability Index)

For Global-X Direction

Level	Load Name	Storey Height (m)	Gravity Load P (kN)	Relative Displacements (mm)	Storey Shear (kN)	Stability Index	Sway Condition

		A	B	C	D	B x C / (A x D)	
0 m to 1.981 m	LOAD 1: EQ X	1.981	4411.575	2.668	301.699	0.02	Non Sway
1.981 m to 5.029 m	LOAD 1: EQ X	3.048	3176.866	9.22	295.532	0.033	Non Sway
5.029 m to 8.077 m	LOAD 1: EQ X	3.048	1942.156	8.326	255.457	0.021	Non Sway
8.077 m to 11.125 m	LOAD 1: EQ X	3.048	707.447	5.255	152.086	0.008	Non Sway
11.125 m to 13.625 m	LOAD 1: EQ X	2.5	43.983	3.297	10.362	0.006	Non Sway
13.625 m to 14.125 m	LOAD 1: EQ X	0.5	35.965	1.39	10.419	0.01	Non Sway

For Global-Y Direction

Level	Load Name	Storey Height (m)	Gravity Load P (kN)	Relative Displacements (mm)	Storey Shear (kN)	Stability Index	Sway Condition
		A	B	C	D	B x C / (A x D)	
0 m to 1.981 m	LOAD 2: EQ Z	1.981	4411.575	1.317	301.699	0.01	Non Sway
1.981 m to 5.029 m	LOAD 2: EQ Z	3.048	3176.866	4.557	295.532	0.016	Non Sway
5.029 m to 8.077 m	LOAD 2: EQ Z	3.048	1942.156	4.41	255.457	0.011	Non Sway
8.077 m to 11.125 m	LOAD 2: EQ Z	3.048	707.447	2.984	152.086	0.005	Non Sway
11.125 m to 13.625 m	LOAD 2: EQ Z	2.5	43.983	0.343	10.362	0.001	Non Sway
13.625 m to 14.125 m	LOAD 2: EQ Z	0.5	35.965	2.833	10.419	0.02	Non Sway

General Data

Column No.	:	C1	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm

Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	9	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	36.79	kN
Mux	=	3.3	kNm
Muy	=	36	kNm
Vux	=	26.82	kN
Vuy	=	2.84	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	No Beam	4991 x 230 x 430	-	30.533	0.827

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.93

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	

Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	3924 x 230 x 430	No Beam	38.835	-	0.495

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.82

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity = Unsupported Length / 500 + D / 30

= 18.1 mm

Mminx = Pu x Minimum Eccentricity

= 0.74 kNm

Slenderness Check

Max Slenderness Ratio(L/B) = 6.74

< 60 (Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor = 0.93

Slenderness Ratio = Effective Length / D

= 3.21, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor = 0.82

Slenderness Ratio = Effective Length / B

= 5.53, Column Not Slender Along B

Calculation of Design Moment

Direction	**Manalysis**	**Mmin** (Abs)	**Mdesign**	**Mslndx** (Abs)	**Mdesign-final**
	A	B	C	E	F
Major Axis - Mux	3.3	0.74	3.3	0	3.3
Minor Axis - Muy	36	---	36	0	36

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	36.79	kN
Mux	=	3.3	kNm
Muy	=	36	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	84.76	deg
MRes	=	36.15	kNm
MCap	=	47.39	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.76 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	29.0895	26.8192
Pu (kN)	211.28	36.79
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3146	0.3275
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2742	1.0539
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2742	1.0539
Enhanced shear strength, Tc-e (N/sqmm)	0.6628	0.5482
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor

Shear Links Design	Shear Links Not Required	Shear Links Not Required

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	Shear Design	Ductile Design	Normal Zone	Ductile Zone
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C1	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	

D/B Ratio	:	1.96 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	12	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	24.25	kN
Mux	=	-3.29	kNm
Muy	=	41.65	kNm
Vux	=	26.53	kN
Vuy	=	2.71	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.827
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.86

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.495
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.67

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed
Most critical case is with Min. Eccentricity check in X-direction
Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.49	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D
Slenderness Check Along D:

Effective Length Factor	=	0.86
Slenderness Ratio	=	Effective Length / D
	=	5, Column not Slender Along D

Column Is Unbraced Along B
Slenderness Check Along B:

Effective Length Factor	=	0.67
Slenderness Ratio	=	Effective Length / B
	=	7.63, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-3.29	0.49	-3.29	0	-3.29
Minor Axis - Muy	41.65	---	41.65	0	41.65

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	24.25	kN
Mux	=	-3.29	kNm
Muy	=	41.65	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	85.49	deg
MRes	=	41.77	kNm
MCap	=	46.5	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.9 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	27.3791	26.5319
Pu (kN)	149.96	24.25
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2961	0.324
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1946	1.0355
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1946	1.0355
Enhanced shear strength, Tc-e (N/sqmm)	0.6214	0.5387
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	

	=	4	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C1	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	15
Critical Analysis Load Combination	:	19
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5

		(LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	95.43	kN
Mux	=	11.78	kNm
Muy	=	39.74	kNm
Vux	=	-25.02	kN
Vuy	=	6.96	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.85

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.65

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	

	=	1.93	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)
Column Is Unbraced Along D			
Slenderness Check Along D:			
Effective Length Factor	=	0.85	
Slenderness Ratio	=	Effective Length / D	
	=	4.95, Column not Slender Along D	
Column Is Unbraced Along B			
Slenderness Check Along B:			
Effective Length Factor	=	0.65	
Slenderness Ratio	=	Effective Length / B	
	=	7.4, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	11.78	1.93	11.78	0	11.78
Minor Axis - Muy	39.74	---	39.74	0	39.74

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	95.43	kN
Mux	=	11.78	kNm
Muy	=	39.74	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	73.49	deg
MRes	=	41.45	kNm
MCap	=	52.32	kNm

Capacity Ratio	=	MRes/ MCap
	=	0.79 <= 1

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	15	17
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	27.2188	22.2093
Pu (kN)	122.02	17.97
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2944	0.2712
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1584	1.0263
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1584	1.0263
Enhanced shear strength, Tc-e (N/sqmm)	0.6026	0.5339
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C1	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	18	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	5.6	kN
Mux	=	0.45	kNm
Muy	=	-26.09	kNm
Vux	=	15.47	kN
Vuy	=	1.29	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.652

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.81

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.278

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.62

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.11	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.81

Slenderness Ratio	=	Effective Length / D
	=	4.71, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.62
Slenderness Ratio	=	Effective Length / B
	=	7.06, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	0.45	0.11	0.45	0	0.45
Minor Axis - Muy	-26.09	---	-26.09	0	-26.09

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	5.6	kN
Mux	=	0.45	kNm
Muy	=	-26.09	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	89	deg
MRes	=	26.09	kNm
MCap	=	44.68	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.58 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	15	17
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)

Design shear force, Vu (kN)	13.9243	15.4716
Pu (kN)	36.95	5.6
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.1506	0.1889
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.048	1.0082
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.048	1.0082
Enhanced shear strength, Tc-e (N/sqmm)	0.5451	0.5245
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C2	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	10	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	281.85	kN
Mux	=	2.27	kNm
Muy	=	48.38	kNm
Vux	=	45.08	kN
Vuy	=	2.77	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	

Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	No Beam	4991 x 230 x 430	-	30.533	0.827

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.93

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.329

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.77

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity = Unsupported Length / 500 + D / 30

= 18.1 mm

Mminx = Pu x Minimum Eccentricity

= 5.64 kNm

Slenderness Check

Max Slenderness Ratio(L/B) = 6.74

< 60 (Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor = 0.93

Slenderness Ratio = Effective Length / D

= 3.21, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor = 0.77

Slenderness Ratio = Effective Length / B

= 5.19, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	2.27	5.64	5.64	0	5.64
Minor Axis - Muy	48.38	---	48.38	0	48.38

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	281.85	kN
Mux	=	5.64	kNm
Muy	=	48.38	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	83.35	deg
MRes	=	48.71	kNm
MCap	=	76.71	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.63 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-25.9716	43.4605
Pu (kN)	103.68	166.88
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2823	0.5365
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength	1 + 3 x Pu / (B x D x Fck)	

Enhancement Factor (Compressive Force)		
	1.1352	1.2472
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1352	1.2472
Enhanced shear strength, Tc-e (N/sqmm)	0.727	0.7987
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C2	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm

Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	13	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	188.96	kN
Mux	=	-2.98	kNm
Muy	=	67.41	kNm
Vux	=	43.95	kN
Vuy	=	2.46	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.827
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.86

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.329
Top	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.6	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	3.82	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.86	
Slenderness Ratio	=	Effective Length / D	
	=	5, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.6	
Slenderness Ratio	=	Effective Length / B	
	=	6.83, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-2.98	3.82	-3.82	0	-3.82
Minor Axis - Muy	67.41	---	67.41	0	67.41

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	188.96	kN
Mux	=	-3.82	kNm
Muy	=	67.41	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	86.75	deg
MRes	=	67.52	kNm
MCap	=	78.17	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.86 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-24.5889	42.4766
Pu (kN)	65.77	111.18
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2673	0.5244
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0858	1.1647
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5

Shear Strength Enhancement Factor	1.0858	1.1647
Enhanced shear strength, Tc-e (N/sqmm)	0.6953	0.7458
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C2	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm

Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	16	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	96.83	kN
Mux	=	6.53	kNm
Muy	=	-58.91	kNm
Vux	=	38.65	kN
Vuy	=	3.3	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.85

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278
Top	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.59	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	1.96	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.85	
Slenderness Ratio	=	Effective Length / D	
	=	4.95, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.59	
Slenderness Ratio	=	Effective Length / B	
	=	6.72, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	6.53	1.96	6.53	0	6.53
Minor Axis - Muy	-58.91	---	-58.91	0	-58.91

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	96.83	kN
Mux	=	6.53	kNm
Muy	=	-58.91	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	83.68	deg
MRes	=	59.27	kNm
MCap	=	73.99	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.8 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	23.5798	37.3823
Pu (kN)	88.94	55.93
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2563	0.4615
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.116	1.0829
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.116	1.0829
Enhanced shear strength, Tc-e (N/sqmm)	0.7146	0.6934
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C2	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	19	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	2.02	kN
Mux	=	-8.52	kNm
Muy	=	-52.03	kNm
Vux	=	32.01	kN
Vuy	=	-3.77	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.652

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.81

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278
Top	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.162

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.57

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.04	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.81	
Slenderness Ratio	=	Effective Length / D	
	=	4.71, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.57	
Slenderness Ratio	=	Effective Length / B	
	=	6.49, Column Not Slender Along B	

Calculation of Design Moment

Direction	**Manalysis**	**Mmin** (Abs)	**Mdesign**	**Mslndx** (Abs)	**Mdesign-final**
	A	B	C	E	F
Major Axis - Mux	-8.52	0.04	-8.52	0	-8.52
Minor Axis - Muy	-52.03	---	-52.03	0	-52.03

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	2.02	kN
Mux	=	-8.52	kNm
Muy	=	-52.03	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	80.71	deg
MRes	=	52.72	kNm
MCap	=	70.16	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.75 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-15.916	32.0076
Pu (kN)	-0.77	2.02
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.173	0.3952
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	-	1.003
Shear Strength Enhancement Factor (Tensile Force)	1	-
Shear Strength Enhancement Factor (max)	-	1.5
Shear Strength Enhancement Factor	-	1.003
Enhanced shear strength, Tc-e (N/sqmm)	0.6404	0.6423
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C3	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	11
Critical Analysis Load Combination	:	12
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Critical Location	=	Bottom Joint

Pu	=	558.79	kN
Mux	=	-5.2	kNm
Muy	=	37.46	kNm
Vux	=	28.23	kN
Vuy	=	11.49	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	No Beam	4991 x 230 x 430	-	30.533	0.827

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.93

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.74

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	11.18	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.93
Slenderness Ratio	=	Effective Length / D
	=	3.21, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.74
Slenderness Ratio	=	Effective Length / B
	=	4.99, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-5.2	11.18	-11.18	0	-11.18
Minor Axis - Muy	37.46	---	37.46	0	37.46

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	558.79	kN
Mux	=	-11.18	kNm
Muy	=	37.46	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	73.39	deg
MRes	=	39.1	kNm
MCap	=	56.7	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.69 <= 1	

Shear Calculation	Along D	Along B

Critical Analysis Load Combination	20	19
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	40.8964	-28.3197
Pu (kN)	360.38	158.02
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4423	0.3458
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.4677	1.2315
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.4677	1.2315
Enhanced shear strength, Tc-e (N/sqmm)	0.7635	0.6406
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			**Provided**	

	Normal Design	Shear Design	Ductile Design	Normal Zone	Ductile Zone
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C3	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	14	
Critical Analysis Load Combination	:	19	
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	125.02	kN
Mux	=	-4.07	kNm
Muy	=	-43.36	kNm
Vux	=	-27.8	kN
Vuy	=	3.19	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.827
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.86

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.53	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.86	
Slenderness Ratio	=	Effective Length / D	
	=	5, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55
Slenderness Ratio	=	Effective Length / B
	=	6.26, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-4.07	2.53	-4.07	0	-4.07
Minor Axis - Muy	-43.36	---	-43.36	0	-43.36

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	125.02	kN
Mux	=	-4.07	kNm
Muy	=	-43.36	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	84.64	deg
MRes	=	43.55	kNm
MCap	=	52.4	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.83 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	19
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	38.7746	-27.8003
Pu (kN)	275.3	125.02

Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4194	0.3394
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.3573	1.1832
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.3573	1.1832
Enhanced shear strength, Tc-e (N/sqmm)	0.7061	0.6155
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C3	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	17	
Critical Analysis Load Combination	:	19	
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	97.62	kN
Mux	=	6.44	kNm
Muy	=	37.3	kNm
Vux	=	-23.53	kN
Vuy	=	4.1	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79
			4991 x 230 x			

Top	57.302	No Beam	430	-	30.533	0.79

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.85

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	1.98	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.85
Slenderness Ratio	=	Effective Length / D
	=	4.95, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55
Slenderness Ratio	=	Effective Length / B
	=	6.26, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	6.44	1.98	6.44	0	6.44
Minor Axis - Muy	37.3	---	37.3	0	37.3

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	97.62	kN
Mux	=	6.44	kNm
Muy	=	37.3	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	80.2	deg
MRes	=	37.85	kNm
MCap	=	51.28	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.74 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	19
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	33.6386	-23.5253
Pu (kN)	183.65	97.62
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3638	0.2872
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength	1 + 3 x Pu / (B x D x Fck)	

Enhancement Factor (Compressive Force)		
	1.2383	1.143
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2383	1.143
Enhanced shear strength, Tc-e (N/sqmm)	0.6442	0.5946
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C3	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm

Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	20	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	115.66	kN
Mux	=	5.21	kNm
Muy	=	37.3	kNm
Vux	=	-20.01	kN
Vuy	=	4.83	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	No Beam	4991 x 230 x 430	-	30.533	0.79
Top	57.302	No Beam	4991 x 230 x 430	-	30.533	0.93

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.9

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.415

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.6	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.34	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.9	
Slenderness Ratio	=	Effective Length / D	
	=	5.24, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.6	
Slenderness Ratio	=	Effective Length / B	
	=	6.83, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	5.21	2.34	5.21	0	5.21
Minor Axis - Muy	37.3	---	37.3	0	37.3

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	115.66	kN
Mux	=	5.21	kNm
Muy	=	37.3	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	82.05	deg
MRes	=	37.66	kNm
MCap	=	51.96	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.72 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	14
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	20.2457	-20.0118
Pu (kN)	99.3	115.66
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.219	0.2443
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1289	1.1695
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5

Shear Strength Enhancement Factor	1.1289	1.1695
Enhanced shear strength, Tc-e (N/sqmm)	0.5872	0.6083
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C3	
Level	:	13.625 m To 14.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2700	mm

Clear Floor Height @ D	=	2700	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	181	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	4.05	kN
Mux	=	-10.39	kNm
Muy	=	18.49	kNm
Vux	=	8.68	kN
Vuy	=	6.03	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	349.312	No Beam	No Beam	-	-	0.93
Top	349.312	No Beam	4991 x 230 x 300	-	10.369	0.971

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.96

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	91.252	No Beam	No Beam	-	-	0.415
Top	91.252	1372 x 230 x 300	No Beam	37.719	-	0.708

Sway Condition (as per Stability	=	Non Sway

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Effective Length Factor along Minor axis	=	0.72	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.4	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.08	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.96	
Slenderness Ratio	=	Effective Length / D	
	=	5.76, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.72	
Slenderness Ratio	=	Effective Length / B	
	=	8.45, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-10.39	0.08	-10.39	0	-10.39
Minor Axis - Muy	18.49	---	18.49	0	18.49

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	4.05	kN

Mux	=	-10.39	kNm
Muy	=	18.49	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	60.68	deg
MRes	=	21.21	kNm
MCap	=	57.58	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.37 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	15	12
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	8.9701	8.6835
Pu (kN)	15.62	4.05
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.097	0.106
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0203	1.0059
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0203	1.0059
Enhanced shear strength, Tc-e (N/sqmm)	0.5307	0.5233
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C4	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	29	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	102.8	kN
Mux	=	8.69	kNm
Muy	=	40.62	kNm
Vux	=	31.23	kN
Vuy	=	-5.96	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.706

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.89

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	3924 x 230 x 430	No Beam	38.835	-	0.495

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.82

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30

	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.06	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.89	
Slenderness Ratio	=	Effective Length / D	
	=	3.07, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.82	
Slenderness Ratio	=	Effective Length / B	
	=	5.53, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	8.69	2.06	8.69	0	8.69
Minor Axis - Muy	40.62	---	40.62	0	40.62

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	102.8	kN
Mux	=	8.69	kNm
Muy	=	40.62	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	77.93	deg

MRes	=	41.53	kNm
MCap	=	51.57	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.81 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-37.8614	31.2325
Pu (kN)	215.96	102.8
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4095	0.3813
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2803	1.1506
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2803	1.1506
Enhanced shear strength, Tc-e (N/sqmm)	0.666	0.5985
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm

Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C4	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	32	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	70.52	kN
Mux	=	9.35	kNm
Muy	=	48.15	kNm
Vux	=	30.78	kN

Vuy	=	-5.71	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.706
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.79

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.495
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.67

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	1.43	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.79
Slenderness Ratio	=	Effective Length / D
	=	4.6, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.67
Slenderness Ratio	=	Effective Length / B
	=	7.63, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	9.35	1.43	9.35	0	9.35
Minor Axis - Muy	48.15	---	48.15	0	48.15

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	70.52	kN
Mux	=	9.35	kNm
Muy	=	48.15	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	79.01	deg
MRes	=	49.05	kNm
MCap	=	50.25	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.98 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17

Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-37.9377	30.7756
Pu (kN)	150.95	70.52
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4103	0.3758
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1959	1.1033
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1959	1.1033
Enhanced shear strength, Tc-e (N/sqmm)	0.6221	0.5739
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---

Spacing	225	---	---	225	---

General Data

Column No.	:	C4	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	35	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	44.8	kN
Mux	=	-8.01	kNm
Muy	=	-41.85	kNm
Vux	=	26.41	kN
Vuy	=	-5.36	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column	Beam Sizes		Beam Stiffness		Beta
		Beam 1	Beam 2			

	Stiffness	(Length x Width x Depth)	(Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.77

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.65

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.91	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.77
Slenderness Ratio	=	Effective Length / D
	=	4.48, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.65

Slenderness Ratio	=	Effective Length / B
	=	7.4, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-8.01	0.91	-8.01	0	-8.01
Minor Axis - Muy	-41.85	---	-41.85	0	-41.85

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	44.8	kN
Mux	=	-8.01	kNm
Muy	=	-41.85	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	79.17	deg
MRes	=	42.61	kNm
MCap	=	49.01	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.87 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-33.9734	26.4108
Pu (kN)	86.19	44.8
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900

Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3674	0.3225
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1119	1.0656
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1119	1.0656
Enhanced shear strength, Tc-e (N/sqmm)	0.5784	0.5543
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C4
Level	:	8.077 m To 11.125 m

Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	38	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	27.48	kN
Mux	=	-16.88	kNm
Muy	=	-33.15	kNm
Vux	=	19.67	kN
Vuy	=	-10.49	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.486

Sway Condition (as per Stability

Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.72

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.278

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.62	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.56	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.72	
Slenderness Ratio	=	Effective Length / D	
	=	4.19, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.62	
Slenderness Ratio	=	Effective Length / B	
	=	7.06, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F

Major Axis - Mux	-16.88	0.56	-16.88	0	-16.88
Minor Axis - Muy	-33.15	---	-33.15	0	-33.15

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	27.48	kN
Mux	=	-16.88	kNm
Muy	=	-33.15	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	6-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	63.01	deg
MRes	=	37.2	kNm
MCap	=	56.6	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.66 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-29.2508	19.6684
Pu (kN)	39.64	27.48
Deff , Beff (mm)	402	182
Aeff (sqmm)	92460	81900
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3164	0.2402
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0514	1.0403

Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0514	1.0403
Enhanced shear strength, Tc-e (N/sqmm)	0.547	0.5411
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C5	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm

Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	30	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	222.15	kN
Mux	=	6.94	kNm
Muy	=	53.22	kNm
Vux	=	49.78	kN
Vuy	=	-4.87	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.706

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.89

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	

Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.329

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.77	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	4.44	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.89
Slenderness Ratio	=	Effective Length / D
	=	3.07, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.77
Slenderness Ratio	=	Effective Length / B
	=	5.19, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	6.94	4.44	6.94	0	6.94
Minor Axis - Muy	53.22	---	53.22	0	53.22

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	222.15	kN
Mux	=	6.94	kNm
Muy	=	53.22	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	82.57	deg
MRes	=	53.67	kNm
MCap	=	75.93	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.71 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-43.7193	47.8286
Pu (kN)	139.32	131.59
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4752	0.5905
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1817	1.1949
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1817	1.1949
Enhanced shear strength, Tc-e (N/sqmm)	0.7567	0.7652
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor

Shear Links Design	Shear Links Not Required	Shear Links Not Required

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	Shear Design	Ductile Design	Normal Zone	Ductile Zone
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C5	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	

D/B Ratio	:	1.96 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	33	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	143.22	kN
Mux	=	7.97	kNm
Muy	=	74.16	kNm
Vux	=	48.34	kN
Vuy	=	-4.82	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.706
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.79

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.329
Top	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.6

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed
Most critical case is with Min. Eccentricity check in X-direction
Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.9	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D
Slenderness Check Along D:

Effective Length Factor	=	0.79
Slenderness Ratio	=	Effective Length / D
	=	4.6, Column not Slender Along D

Column Is Unbraced Along B
Slenderness Check Along B:

Effective Length Factor	=	0.6
Slenderness Ratio	=	Effective Length / B
	=	6.83, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	7.97	2.9	7.97	0	7.97
Minor Axis - Muy	74.16	---	74.16	0	74.16

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	143.22	kN
Mux	=	7.97	kNm
Muy	=	74.16	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	83.86	deg
MRes	=	74.59	kNm
MCap	=	75.29	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.99 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-44.1235	46.6493
Pu (kN)	91.05	84.19
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4796	0.5759
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1188	1.1247
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1188	1.1247
Enhanced shear strength, Tc-e (N/sqmm)	0.7164	0.7202
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	

	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C5	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	36
Critical Analysis Load Combination	:	12
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5

		(LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	62.14	kN
Mux	=	8.83	kNm
Muy	=	65.06	kNm
Vux	=	42.81	kN
Vuy	=	-5.29	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.77

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278
Top	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.59

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	

	=	1.26	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)
Column Is Unbraced Along D			
Slenderness Check Along D:			
Effective Length Factor	=	0.77	
Slenderness Ratio	=	Effective Length / D	
	=	4.48, Column not Slender Along D	
Column Is Unbraced Along B			
Slenderness Check Along B:			
Effective Length Factor	=	0.59	
Slenderness Ratio	=	Effective Length / B	
	=	6.72, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	8.83	1.26	8.83	0	8.83
Minor Axis - Muy	65.06	---	65.06	0	65.06

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	62.14	kN
Mux	=	8.83	kNm
Muy	=	65.06	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	82.27	deg
MRes	=	65.66	kNm
MCap	=	72.45	kNm

Capacity Ratio	=	MRes/ MCap
	=	0.91 <= 1

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	13	17
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-42.4936	41.294
Pu (kN)	67.83	35.48
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4619	0.5098
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0885	1.0526
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0885	1.0526
Enhanced shear strength, Tc-e (N/sqmm)	0.697	0.674
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C5	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	39	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	-23.89	kN
Mux	=	-16.42	kNm
Muy	=	-54.05	kNm
Vux	=	33.38	kN
Vuy	=	-10.39	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.486

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.72

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.278
Top	14.969	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.162

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.57

Minimum Eccentricity Check

Since Axial Force is tensile, Min. Eccentricity check not to be performed

Slenderness Check

Since Axial Force is tensile, slenderness check not to be performed

Check for Modulus of Rupture

Axial Tensile Stress	=	Pu / (B x D)	
	=	0.2308	N/sqmm
Modulus of Rupture (Fcr)	=	0.7 x Square Root (Fck)	
	=	3.5	N/sqmm
Since,Axial tensile stress in column	<	Fcr	

Final Critical Design Forces

Pu	=	-23.89	kN
Mux	=	16.42	kNm
Muy	=	54.05	kNm

Resultant Based

Moment Capacity Check

Pt Calculated	=	1.99	
Reinforcement Provided	=	4-T20 + 4-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	73.1	deg
MRes	=	56.49	kNm
MCap	=	70.48	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.8 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-34.9953	33.3783
Pu (kN)	-18.04	-23.89
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3804	0.4121
Pt (%)	0.996	0.996
Design shear strength, Tc (N/sqmm)	0.6404	0.6404
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	-	-
Shear Strength Enhancement Factor (Tensile Force)	1	1
Shear Strength Enhancement Factor (max)	-	-
Shear Strength Enhancement Factor	-	-
Enhanced shear strength, Tc-e (N/sqmm)	0.6404	0.6404
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C6	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	31	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	491.59	kN
Mux	=	11.93	kNm
Muy	=	-42.79	kNm
Vux	=	-38.23	kN
Vuy	=	-8.49	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.706

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.89

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.74

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	9.83	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.89	
Slenderness Ratio	=	Effective Length / D	
	=	3.07, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.74	
Slenderness Ratio	=	Effective Length / B	
	=	4.99, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	11.93	9.83	11.93	0	11.93
Minor Axis - Muy	-42.79	---	-42.79	0	-42.79

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	491.59	kN
Mux	=	11.93	kNm
Muy	=	-42.79	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	

Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	74.42	deg
MRes	=	44.43	kNm
MCap	=	66.62	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.67 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	19
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-53.1582	-35.4691
Pu (kN)	382.34	250.72
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.5778	0.4379
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.4987	1.3714
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.4987	1.3714
Enhanced shear strength, Tc-e (N/sqmm)	0.8838	0.8088
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm

Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C6	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	34	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	364.95	kN
Mux	=	13.26	kNm

Muy	=	-57.5	kNm
Vux	=	-37.16	kN
Vuy	=	-8.21	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.706
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.79

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	7.39	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38

	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.79	
Slenderness Ratio	=	Effective Length / D	
	=	4.6, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55	
Slenderness Ratio	=	Effective Length / B	
	=	6.26, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	13.26	7.39	13.26	0	13.26
Minor Axis - Muy	-57.5	---	-57.5	0	-57.5

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	364.95	kN
Mux	=	13.26	kNm
Muy	=	-57.5	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	77.01	deg
MRes	=	59.01	kNm
MCap	=	66.1	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.89 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	19
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-53.1529	-34.5763
Pu (kN)	283.04	187.24
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.5777	0.4269
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.3692	1.2774
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.3692	1.2774
Enhanced shear strength, Tc-e (N/sqmm)	0.8074	0.7533
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C6	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	37	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	245.62	kN
Mux	=	-11.35	kNm
Muy	=	51.3	kNm
Vux	=	-32.66	kN
Vuy	=	-7.73	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654
Top	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.77

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	4.97	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.77	
Slenderness Ratio	=	Effective Length / D	
	=	4.48, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55
Slenderness Ratio	=	Effective Length / B
	=	6.26, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-11.35	4.97	-11.35	0	-11.35
Minor Axis - Muy	51.3	---	51.3	0	51.3

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	245.62	kN
Mux	=	-11.35	kNm
Muy	=	51.3	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	77.52	deg
MRes	=	52.54	kNm
MCap	=	65.33	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.8 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	19
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-48.4966	-30.1753

Pu (kN)	183.7	130.38
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.5271	0.3725
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2396	1.1932
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2396	1.1932
Enhanced shear strength, Tc-e (N/sqmm)	0.731	0.7036
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C6	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	40	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	119.41	kN
Mux	=	-21.85	kNm
Muy	=	48.87	kNm
Vux	=	-27.57	kN
Vuy	=	-12.48	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4991 x 230 x 430	5067 x 230 x 430	30.533	30.075	0.654
		4991 x 230 x	5067 x 230 x			

Top	57.302	430	430	30.533	30.075	0.87

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.83

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.415

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.6

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.42	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.83
Slenderness Ratio	=	Effective Length / D
	=	4.83, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.6
Slenderness Ratio	=	Effective Length / B
	=	6.83, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-21.85	2.42	-21.85	0	-21.85
Minor Axis - Muy	48.87	---	48.87	0	48.87

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	119.41	kN
Mux	=	-21.85	kNm
Muy	=	48.87	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	65.91	deg
MRes	=	53.53	kNm
MCap	=	70.43	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.76 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	13	14
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-39.7544	-27.5717
Pu (kN)	142.23	119.41
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4321	0.3404
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength	1 + 3 x Pu / (B x D x Fck)	

Enhancement Factor (Compressive Force)		
	1.1855	1.1769
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1855	1.1769
Enhanced shear strength, Tc-e (N/sqmm)	0.6991	0.694
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C6	
Level	:	13.625 m To 14.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm

Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2700	mm
Clear Floor Height @ D	=	2700	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	182	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	8.15	kN
Mux	=	12.54	kNm
Muy	=	17.32	kNm
Vux	=	7.85	kN
Vuy	=	-7.65	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	349.312	No Beam	No Beam	-	-	0.87
Top	349.312	4991 x 230 x 300	No Beam	10.369	-	0.971

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.94

Calculation Along Minor Axis Of Column

		Beam Sizes		Beam Stiffness		

Joint	Column Stiffness	Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	Beta
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	91.252	No Beam	No Beam	-	-	0.415
Top	91.252	1372 x 230 x 300	No Beam	37.719	-	0.708

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.72

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.4	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.17	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.94	
Slenderness Ratio	=	Effective Length / D	
	=	5.64, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.72	
Slenderness Ratio	=	Effective Length / B	
	=	8.45, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	12.54	0.17	12.54	0	12.54
Minor Axis - Muy	17.32	---	17.32	0	17.32

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	8.15	kN
Mux	=	12.54	kNm
Muy	=	17.32	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	54.09	deg
MRes	=	21.38	kNm
MCap	=	83.06	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.26 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-13.5262	7.8469
Pu (kN)	20.16	8.15
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.147	0.0969
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0263	1.0121
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0263	1.0121
Enhanced shear	0.6052	0.5968

strength, Tc-e (N/sqmm)		
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C7	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	

Column Type	:	UnBraced
Minimum eccentricity check	:	One Axis at a Time
Code defined D/B ratio	:	4
D/B Ratio	:	1.96 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	49	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	72.37	kN
Mux	=	9.47	kNm
Muy	=	42.77	kNm
Vux	=	33.23	kN
Vuy	=	-7.24	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.695

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.89

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	3924 x 230 x 430	No Beam	38.835	-	0.495

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.82

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	1.45	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.89	
Slenderness Ratio	=	Effective Length / D	
	=	3.07, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.82	
Slenderness Ratio	=	Effective Length / B	
	=	5.53, Column Not Slender Along B	

Calculation of Design Moment

Direction	**Manalysis**	**Mmin** (Abs)	**Mdesign**	**Mslndx** (Abs)	**Mdesign-final**
	A	B	C	E	F
Major Axis - Mux	9.47	1.45	9.47	0	9.47
Minor Axis - Muy	42.77	---	42.77	0	42.77

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	72.37	kN
Mux	=	9.47	kNm
Muy	=	42.77	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	77.51	deg
MRes	=	43.8	kNm
MCap	=	62.06	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.71 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-39.9557	33.2268
Pu (kN)	170.31	72.37
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4343	0.4102
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2221	1.1072
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2221	1.1072
Enhanced shear strength, Tc-e (N/sqmm)	0.7207	0.6529
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm

	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C7	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	52
Critical Analysis Load Combination	:	17

Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	48.32	kN
Mux	=	11.44	kNm
Muy	=	50.93	kNm
Vux	=	32.63	kN
Vuy	=	-7.17	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.695
Top	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.78

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.495
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.67

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm

Mminx	=	Pu x Minimum Eccentricity	
	=	0.98	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.78	
Slenderness Ratio	=	Effective Length / D	
	=	4.54, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.67	
Slenderness Ratio	=	Effective Length / B	
	=	7.63, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	11.44	0.98	11.44	0	11.44
Minor Axis - Muy	50.93	---	50.93	0	50.93

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	48.32	kN
Mux	=	11.44	kNm
Muy	=	50.93	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	77.34	deg
MRes	=	52.2	kNm

MCap	=	61.46	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.85 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-40.3691	32.6261
Pu (kN)	116	48.32
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4388	0.4028
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1513	1.0716
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1513	1.0716
Enhanced shear strength, Tc-e (N/sqmm)	0.6789	0.6319
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C7	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	55	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	31.09	kN
Mux	=	-8.45	kNm
Muy	=	-43.95	kNm
Vux	=	27.64	kN
Vuy	=	-5.96	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642
Top	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.76

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.65

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.63	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor = 0.76

Slenderness Ratio	=	Effective Length / D
	=	4.42, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.65
Slenderness Ratio	=	Effective Length / B
	=	7.4, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-8.45	0.63	-8.45	0	-8.45
Minor Axis - Muy	-43.95	---	-43.95	0	-43.95

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	31.09	kN
Mux	=	-8.45	kNm
Muy	=	-43.95	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	79.12	deg
MRes	=	44.75	kNm
MCap	=	60.68	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.74 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ

	Z)	X)
Design shear force, Vu (kN)	-35.0148	27.6435
Pu (kN)	64.09	31.09
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3806	0.3413
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0836	1.0461
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0836	1.0461
Enhanced shear strength, Tc-e (N/sqmm)	0.639	0.6169
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C7	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	58	
Critical Analysis Load Combination	:	13	
Load Combination	=	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Critical Location	=	Top Joint	
Pu	=	18.96	kN
Mux	=	-63.23	kNm
Muy	=	-4.41	kNm
Vux	=	2.38	kN
Vuy	=	-35.31	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	

	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642
Top	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.473

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.72

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.278

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.62

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.38	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.72
Slenderness Ratio	=	Effective Length / D
	=	4.19, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.62
Slenderness Ratio	=	Effective Length / B
	=	7.06, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-63.23	0.38	-63.23	0	-63.23
Minor Axis - Muy	-4.41	---	-4.41	0	-4.41

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	18.96	kN
Mux	=	-63.23	kNm
Muy	=	-4.41	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	3.99	deg
MRes	=	63.39	kNm
MCap	=	138.13	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.46 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-35.3067	17.5586
Pu (kN)	18.96	12.11
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3838	0.2168
Pt (%)	0.801	0.801

Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0247	1.0179
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0247	1.0179
Enhanced shear strength, Tc-e (N/sqmm)	0.6043	0.6003
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C8
Level	:	0 m To 1.981 m
Frame Type	=	Non-Ductile
Response Reduction Factor	=	3
Design Code	=	IS 456 : 2000 + IS 13920 : 2016

Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	50	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	212.1	kN
Mux	=	10.64	kNm
Muy	=	62.93	kNm
Vux	=	57.44	kN
Vuy	=	-7.05	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	144.042	No Beam	No Beam	-	-	1
Top	144.042	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.788

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.92

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	27.126	No Beam	No Beam	-	-	1
Top	27.126	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.366

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.78

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.77	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	4.4	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.92
Slenderness Ratio	=	Effective Length / D
	=	2.69, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.78
Slenderness Ratio	=	Effective Length / B
	=	5.26, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	10.64	4.4	10.64	0	10.64
Minor Axis - Muy	62.93	---	62.93	0	62.93

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	212.1	kN
Mux	=	10.64	kNm
Muy	=	62.93	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	80.4	deg
MRes	=	63.82	kNm
MCap	=	97.63	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.65 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-55.6685	55.2971
Pu (kN)	99.17	127.11
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.5042	0.5796
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1078	1.1599
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5

Shear Strength Enhancement Factor	1.1078	1.1599
Enhanced shear strength, Tc-e (N/sqmm)	0.7588	0.7945
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C8	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm

Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	53	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	137.33	kN
Mux	=	11.9	kNm
Muy	=	85.68	kNm
Vux	=	55.77	kN
Vuy	=	-7.05	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	93.618	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.788
Top	93.618	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.745

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.84

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.366
Top	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.61	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	22.9	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	3.15	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.84	
Slenderness Ratio	=	Effective Length / D	
	=	4.15, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.61	
Slenderness Ratio	=	Effective Length / B	
	=	6.94, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	11.9	3.15	11.9	0	11.9
Minor Axis - Muy	85.68	---	85.68	0	85.68

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	137.33	kN
Mux	=	11.9	kNm
Muy	=	85.68	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	82.09	deg
MRes	=	86.51	kNm
MCap	=	97.77	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.88 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	17
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-55.6825	53.8592
Pu (kN)	61.05	82.22
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.5044	0.5646
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0664	1.1034
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0664	1.1034
Enhanced shear strength, Tc-e (N/sqmm)	0.7304	0.7558
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C8	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	56	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	60.76	kN
Mux	=	8.94	kNm
Muy	=	73.82	kNm
Vux	=	49.04	kN
Vuy	=	-4.05	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	93.618	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.745
Top	93.618	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.745

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.83

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312
Top	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.6

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	22.9	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	1.39	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.83	
Slenderness Ratio	=	Effective Length / D	
	=	4.1, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.6	
Slenderness Ratio	=	Effective Length / B	
	=	6.83, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	8.94	1.39	8.94	0	8.94
Minor Axis - Muy	73.82	---	73.82	0	73.82

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	60.76	kN
Mux	=	8.94	kNm
Muy	=	73.82	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	83.09	deg
MRes	=	74.36	kNm
MCap	=	96.82	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.77 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	12
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-46.2319	49.0435
Pu (kN)	24.41	60.76
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4188	0.5141
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0265	1.0764
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0265	1.0764
Enhanced shear strength, Tc-e (N/sqmm)	0.7032	0.7373
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C8	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	59
Critical Analysis Load Combination	:	12
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Critical Location	=	Top Joint

Pu	=	-20.13	kN
Mux	=	-26.67	kNm
Muy	=	-51.33	kNm
Vux	=	31.8	kN
Vuy	=	-15.48	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	93.618	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.745
Top	93.618	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.594

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.78

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312
Top	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.185

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.58

Minimum Eccentricity Check

Since Axial Force is tensile, Min. Eccentricity check not to be performed

Slenderness Check

Since Axial Force is tensile, slenderness check not to be performed

Check for Modulus of Rupture

Axial Tensile Stress	=	Pu / (B x D)

	=	0.1652	N/sqmm
Modulus of Rupture (Fcr)	=	0.7 x Square Root (Fck)	
	=	3.5	N/sqmm
Since,Axial tensile stress in column	<	Fcr	

Final Critical Design Forces

Pu	=	-20.13	kN
Mux	=	26.67	kNm
Muy	=	51.33	kNm

Resultant Based

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	62.54	deg
MRes	=	57.84	kNm
MCap	=	113.57	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.51 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-44.0826	31.804
Pu (kN)	-25.55	-20.13
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3993	0.3334
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	-	-
Shear Strength Enhancement Factor (Tensile Force)	1	1
Shear Strength Enhancement Factor (max)	-	-
Shear Strength	-	-

Enhancement Factor		
Enhanced shear strength, Tc-e (N/sqmm)	0.685	0.685
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C9	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm

No Of Floors	=	1
No Of Columns In Group	=	1
Column Type	:	UnBraced
Minimum eccentricity check	:	One Axis at a Time
Code defined D/B ratio	:	4
D/B Ratio	:	1.96 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	51	
Critical Analysis Load Combination	:	19	
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	157.71	kN
Mux	=	9.81	kNm
Muy	=	-42.96	kNm
Vux	=	-36.29	kN
Vuy	=	-7.8	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.695

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.89

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202

Sway Condition (as per Stability Index)	=	Non Sway

Effective Length Factor along Minor axis	=	0.74	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	3.15	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.89	
Slenderness Ratio	=	Effective Length / D	
	=	3.07, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.74	
Slenderness Ratio	=	Effective Length / B	
	=	4.99, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	9.81	3.15	9.81	0	9.81
Minor Axis - Muy	-42.96	---	-42.96	0	-42.96

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	157.71	kN
Mux	=	9.81	kNm
Muy	=	-42.96	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	77.14	deg
MRes	=	44.07	kNm
MCap	=	64.16	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.69 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	18	19
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-55.78	-36.2873
Pu (kN)	242.43	157.71
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.6063	0.448
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.3162	1.2337
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.3162	1.2337
Enhanced shear strength, Tc-e (N/sqmm)	0.7762	0.7275
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links			
Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C9	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	54	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	236.62	kN
Mux	=	17.11	kNm
Muy	=	-57.61	kNm
Vux	=	-37.23	kN
Vuy	=	-10.8	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.695
Top	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.78

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30

	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	4.79	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.78	
Slenderness Ratio	=	Effective Length / D	
	=	4.54, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55	
Slenderness Ratio	=	Effective Length / B	
	=	6.26, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	17.11	4.79	17.11	0	17.11
Minor Axis - Muy	-57.61	---	-57.61	0	-57.61

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	236.62	kN
Mux	=	17.11	kNm
Muy	=	-57.61	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	

	=	73.46	deg
MRes	=	60.09	kNm
MCap	=	66.52	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.9 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	18	19
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-55.8971	-35.3206
Pu (kN)	167.96	107.88
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.6076	0.4361
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2191	1.1598
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2191	1.1598
Enhanced shear strength, Tc-e (N/sqmm)	0.7189	0.684
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm

Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C9	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	57	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	135.6	kN
Mux	=	-14.26	kNm
Muy	=	50.85	kNm

Vux	=	-32.19	kN
Vuy	=	-9.76	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642
Top	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.76

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.74	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.76
Slenderness Ratio	=	Effective Length / D
	=	4.42, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55
Slenderness Ratio	=	Effective Length / B
	=	6.26, Column Not Slender Along B

Calculation of Design Moment

Direction	**Manalysis**	**Mmin** (Abs)	**Mdesign**	**Mslndx** (Abs)	**Mdesign-final**
	A	B	C	E	F
Major Axis - Mux	-14.26	2.74	-14.26	0	-14.26
Minor Axis - Muy	50.85	---	50.85	0	50.85

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	135.6	kN
Mux	=	-14.26	kNm
Muy	=	50.85	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	74.33	deg
MRes	=	52.82	kNm
MCap	=	64.69	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.82 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load	18	19

Combination		
Critical Load Combination	[14] : 0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-50.4987	-30.2461
Pu (kN)	95.84	63.76
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.5489	0.3734
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.125	1.0945
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.125	1.0945
Enhanced shear strength, Tc-e (N/sqmm)	0.6634	0.6454
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			**Provided**	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**

Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C9	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	60	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	21.67	kN
Mux	=	-23.77	kNm
Muy	=	39.56	kNm
Vux	=	-22.85	kN
Vuy	=	-14.01	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

		Beam Sizes		Beam Stiffness		

Joint	Column Stiffness	Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	Beta
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.642
Top	57.302	5067 x 230 x 430	4496 x 230 x 430	30.075	33.894	0.473

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.72

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.091

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.54

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.44	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.72	
Slenderness Ratio	=	Effective Length / D	
	=	4.19, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.54
Slenderness Ratio	=	Effective Length / B
	=	6.15, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-23.77	0.44	-23.77	0	-23.77
Minor Axis - Muy	39.56	---	39.56	0	39.56

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	21.67	kN
Mux	=	-23.77	kNm
Muy	=	39.56	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	59	deg
MRes	=	46.16	kNm
MCap	=	76.56	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.6 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	13	14
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-43.9899	-22.8515
Pu (kN)	32.7	21.67
Deff , Beff (mm)	400	180

Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4782	0.2821
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0426	1.0321
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0426	1.0321
Enhanced shear strength, Tc-e (N/sqmm)	0.6149	0.6086
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No. : C10

Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	69	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	108.09	kN
Mux	=	6.69	kNm
Muy	=	44.81	kNm
Vux	=	34.78	kN
Vuy	=	-2.71	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.58

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.84

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	3924 x 230 x 430	No Beam	38.835	-	0.495

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.82

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.16	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.84
Slenderness Ratio	=	Effective Length / D
	=	2.9, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.82
Slenderness Ratio	=	Effective Length / B
	=	5.53, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	6.69	2.16	6.69	0	6.69

Minor Axis - Muy	44.81	---	44.81	0	44.81

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	108.09	kN
Mux	=	6.69	kNm
Muy	=	44.81	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	81.51	deg
MRes	=	45.31	kNm
MCap	=	63.25	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.72 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	25.8506	34.7825
Pu (kN)	155.54	108.09
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.281	0.4294
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2029	1.1601

Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2029	1.1601
Enhanced shear strength, Tc-e (N/sqmm)	0.7094	0.6841
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C10	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm

Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	72	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	396.29	kN
Mux	=	-8.73	kNm
Muy	=	-58.6	kNm
Vux	=	-37.7	kN
Vuy	=	6.42	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.58
Top	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.71

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	

	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.495
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.67	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	8.02	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.71	
Slenderness Ratio	=	Effective Length / D	
	=	4.13, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.67	
Slenderness Ratio	=	Effective Length / B	
	=	7.63, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-8.73	8.02	-8.73	0	-8.73
Minor Axis - Muy	-58.6	---	-58.6	0	-58.6

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	396.29	kN
Mux	=	-8.73	kNm
Muy	=	-58.6	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	81.53	deg
MRes	=	59.25	kNm
MCap	=	66.78	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.89 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	24.5992	34.3098
Pu (kN)	116.63	83.39
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2674	0.4236
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1521	1.1235
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1521	1.1235
Enhanced shear	0.6794	0.6626

strength, Tc-e (N/sqmm)		
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C10	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	

Column Type	:	UnBraced
Minimum eccentricity check	:	One Axis at a Time
Code defined D/B ratio	:	4
D/B Ratio	:	1.96 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	75	
Critical Analysis Load Combination	:	14	
Load Combination	=	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	251.59	kN
Mux	=	3.72	kNm
Muy	=	51.69	kNm
Vux	=	-32.76	kN
Vuy	=	3.46	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521
Top	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.7

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor		

axis	=	0.65	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	5.09	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.7	
Slenderness Ratio	=	Effective Length / D	
	=	4.07, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.65	
Slenderness Ratio	=	Effective Length / B	
	=	7.4, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	3.72	5.09	5.09	0	5.09
Minor Axis - Muy	51.69	---	51.69	0	51.69

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	251.59	kN
Mux	=	5.09	kNm
Muy	=	51.69	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	84.37	deg
MRes	=	51.94	kNm
MCap	=	67.37	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.77 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	19.5211	29.2003
Pu (kN)	83.88	65.02
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2122	0.3605
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1094	1.0963
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1094	1.0963
Enhanced shear strength, Tc-e (N/sqmm)	0.6542	0.6465
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links			
Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C10	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	78	
Critical Analysis Load Combination	:	17	
Load Combination	=	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	37.01	kN
Mux	=	20.21	kNm
Muy	=	-32.49	kNm
Vux	=	19	kN
Vuy	=	7.8	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521
Top	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.547

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.71

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	3924 x 230 x 430	No Beam	38.835	-	0.435
Top	14.969	3924 x 230 x 430	No Beam	38.835	-	0.461

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.66

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30

	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.75	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.71	
Slenderness Ratio	=	Effective Length / D	
	=	4.13, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.66	
Slenderness Ratio	=	Effective Length / B	
	=	7.51, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	20.21	0.75	20.21	0	20.21
Minor Axis - Muy	-32.49	---	-32.49	0	-32.49

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	37.01	kN
Mux	=	20.21	kNm
Muy	=	-32.49	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	

	=	58.11	deg
MRes	=	38.26	kNm
MCap	=	78.27	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.49 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	15	17
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	23.4697	18.996
Pu (kN)	84.19	37.01
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2551	0.2345
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1098	1.0548
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1098	1.0548
Enhanced shear strength, Tc-e (N/sqmm)	0.6545	0.622
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm

Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C10	
Level	:	11.125 m To 13.625 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2200	mm
Clear Floor Height @ D	=	2200	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	173	
Critical Analysis Load Combination	:	13	
Load Combination	=	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Critical Location	=	Bottom Joint	
Pu	=	17.65	kN
Mux	=	41.05	kNm
Muy	=	-3.41	kNm

Vux	=	-3.68	kN
Vuy	=	-15.18	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	69.862	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.547
Top	69.862	No Beam	2134 x 230 x 300	-	24.25	0.742

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.77

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	18.25	3924 x 230 x 430	No Beam	38.835	-	0.461
Top	18.25	3924 x 230 x 300	No Beam	13.188	-	0.581

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.7

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	19.4	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.35	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	9.57	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.77
Slenderness Ratio	=	Effective Length / D
	=	3.76, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.7
Slenderness Ratio	=	Effective Length / B
	=	6.7, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	41.05	0.35	41.05	0	41.05
Minor Axis - Muy	-3.41	---	-3.41	0	-3.41

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	17.65	kN
Mux	=	41.05	kNm
Muy	=	-3.41	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	4.76	deg
MRes	=	41.19	kNm
MCap	=	137.64	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.3 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	13	14

Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-15.1767	-7.4702
Pu (kN)	17.65	24.41
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.165	0.0922
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.023	1.0362
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.023	1.0362
Enhanced shear strength, Tc-e (N/sqmm)	0.6033	0.611
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---

Spacing	225	---	---	225	---

General Data

Column No.	:	C11	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	70	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	468.02	kN
Mux	=	0.74	kNm
Muy	=	65.87	kNm
Vux	=	60.11	kN
Vuy	=	9.54	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column	Beam Sizes		Beam Stiffness		Beta
		Beam 1	Beam 2			

	Stiffness	(Length x Width x Depth)	(Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	144.042	No Beam	No Beam	-	-	1
Top	144.042	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.693

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.89

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	27.126	No Beam	No Beam	-	-	1
Top	27.126	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.366

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.78

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.77	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	9.72	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.89
Slenderness Ratio	=	Effective Length / D
	=	2.6, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.78
Slenderness Ratio	=	Effective Length / B

= 5.26, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	0.74	9.72	9.72	0	9.72
Minor Axis - Muy	65.87	---	65.87	0	65.87

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	468.02	kN
Mux	=	9.72	kNm
Muy	=	65.87	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	81.61	deg
MRes	=	66.58	kNm
MCap	=	97.85	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.68 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	44.0985	58.0079
Pu (kN)	186.91	282.61
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3994	0.608

Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2032	1.3555
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2032	1.3555
Enhanced shear strength, Tc-e (N/sqmm)	0.8242	0.9285
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C11
Level	:	1.981 m To 5.029 m
Frame Type	=	Non-Ductile
Response Reduction Factor	=	3

Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	73	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	343.69	kN
Mux	=	17.37	kNm
Muy	=	-88.17	kNm
Vux	=	58.41	kN
Vuy	=	10	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	93.618	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.693
Top	93.618	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.64

Sway Condition (as per Stability Index)	=	Non Sway

Effective Length Factor along Major

Axis = 0.78

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.366
Top	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.61

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	22.9	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	7.87	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.78
Slenderness Ratio	=	Effective Length / D
	=	3.85, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.61
Slenderness Ratio	=	Effective Length / B
	=	6.94, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	17.37	7.87	17.37	0	17.37
Minor Axis - Muy	-88.17	---	-88.17	0	-88.17

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	343.69	kN
Mux	=	17.37	kNm
Muy	=	-88.17	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	78.86	deg
MRes	=	89.86	kNm
MCap	=	97.13	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.93 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	41.7276	56.5413
Pu (kN)	135.12	208.2
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.378	0.5927
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1469	1.2619
Shear Strength Enhancement Factor	-	-

(Tensile Force)		
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1469	1.2619
Enhanced shear strength, Tc-e (N/sqmm)	0.7856	0.8644
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C11	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm

Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	76	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	218.33	kN
Mux	=	-13.49	kNm
Muy	=	78.45	kNm
Vux	=	52.09	kN
Vuy	=	7.47	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	93.618	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.64
Top	93.618	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.64

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.76

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	

Bottom	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312
Top	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.6	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	22.9	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	5	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.76	
Slenderness Ratio	=	Effective Length / D	
	=	3.75, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.6	
Slenderness Ratio	=	Effective Length / B	
	=	6.83, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-13.49	5	-13.49	0	-13.49
Minor Axis - Muy	78.45	---	78.45	0	78.45

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	218.33	kN
Mux	=	-13.49	kNm
Muy	=	78.45	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	80.24	deg
MRes	=	79.6	kNm
MCap	=	97.59	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.82 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	20	17
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	33.3131	50.2481
Pu (kN)	90.04	133.22
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3017	0.5267
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0979	1.1676
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0979	1.1676
Enhanced shear strength, Tc-e (N/sqmm)	0.752	0.7998

Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C11	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	

Minimum eccentricity check	:	One Axis at a Time
Code defined D/B ratio	:	4
D/B Ratio	:	2.3 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	79	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	91.31	kN
Mux	=	73.51	kNm
Muy	=	-54.74	kNm
Vux	=	34.53	kN
Vuy	=	32.44	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	93.618	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.64
Top	93.618	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.664

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.77

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.312
Top	17.63	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.335

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.61

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	22.9	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.09	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.77	
Slenderness Ratio	=	Effective Length / D	
	=	3.8, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.61	
Slenderness Ratio	=	Effective Length / B	
	=	6.94, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	73.51	2.09	73.51	0	73.51
Minor Axis - Muy	-54.74	---	-54.74	0	-54.74

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	91.31	kN
Mux	=	73.51	kNm
Muy	=	-54.74	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	36.67	deg
MRes	=	91.65	kNm
MCap	=	179.73	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.51 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	15	17
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[13] : 0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	41.7481	32.3708
Pu (kN)	71.17	57.29
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3782	0.3393
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0774	1.0721
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0774	1.0721
Enhanced shear strength, Tc-e (N/sqmm)	0.738	0.7344
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links			
Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C11	
Level	:	11.125 m To 13.625 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	530	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2200	mm
Clear Floor Height @ D	=	2200	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	2.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	174	
Critical Analysis Load Combination	:	5	
Load Combination	=	[1] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Critical Location	=	Bottom Joint	
Pu	=	32.83	kN
Mux	=	54.82	kNm
Muy	=	6.86	kNm
Vux	=	6.34	kN
Vuy	=	-15.22	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	114.139	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.664
Top	114.139	No Beam	2134 x 230 x 300	-	24.25	0.825

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.82

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	21.495	3925 x 230 x 430	3924 x 230 x 430	38.825	38.835	0.335
Top	21.495	No Beam	3924 x 230 x 300	-	13.188	0.62

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.68

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30

	=	22.07	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.72	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	9.57	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.82	
Slenderness Ratio	=	Effective Length / D	
	=	3.4, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.68	
Slenderness Ratio	=	Effective Length / B	
	=	6.5, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	54.82	0.72	54.82	0	54.82
Minor Axis - Muy	6.86	---	6.86	0	6.86

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	32.83	kN
Mux	=	54.82	kNm
Muy	=	6.86	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	2.39	
Reinforcement Provided	=	8-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	

	=	7.14	deg
MRes	=	55.25	kNm
MCap	=	230.11	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.24 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	12
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-18.0741	10.9659
Pu (kN)	23.59	30.14
Deff , Beff (mm)	480	180
Aeff (sqmm)	110400	95400
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.1637	0.1149
Pt (%)	1.196	1.196
Design shear strength, Tc (N/sqmm)	0.685	0.685
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0256	1.0379
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0256	1.0379
Enhanced shear strength, Tc-e (N/sqmm)	0.7026	0.711
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm

Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C12	
Level	:	0 m To 1.981 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	1551	mm
Clear Floor Height @ D	=	1551	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	71	
Critical Analysis Load Combination	:	12	
Load Combination	=	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	671.08	kN
Mux	=	-3.75	kNm
Muy	=	45.83	kNm
Vux	=	35.44	kN

Vuy	=	9.01	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	88.166	No Beam	No Beam	-	-	1
Top	88.166	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.58

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.84

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	23.032	No Beam	No Beam	-	-	1
Top	23.032	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.74

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	18.1	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	13.42	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	6.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.84
Slenderness Ratio	=	Effective Length / D

	=	2.9, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.74
Slenderness Ratio	=	Effective Length / B
	=	4.99, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-3.75	13.42	-13.42	0	-13.42
Minor Axis - Muy	45.83	---	45.83	0	45.83

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	671.08	kN
Mux	=	-13.42	kNm
Muy	=	45.83	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	73.68	deg
MRes	=	47.75	kNm
MCap	=	63.95	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.75 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	19
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu	41.0586	-38.8782

(kN)		
Pu (kN)	218.92	199.1
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4463	0.48
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.2855	1.295
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.2855	1.295
Enhanced shear strength, Tc-e (N/sqmm)	0.7581	0.7637
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C12	
Level	:	1.981 m To 5.029 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	74	
Critical Analysis Load Combination	:	19	
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Bottom Joint	
Pu	=	148.35	kN
Mux	=	5.29	kNm
Muy	=	-58.98	kNm
Vux	=	-38.04	kN
Vuy	=	-3.04	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4496 x 230 x	2134 x 230 x	33.894	71.41	0.58

		430	430			
Top	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.71

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.202
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.55

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	3	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.71	
Slenderness Ratio	=	Effective Length / D	
	=	4.13, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55	
Slenderness Ratio	=	Effective Length / B	
	=	6.26, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	5.29	3	5.29	0	5.29
Minor Axis - Muy	-58.98	---	-58.98	0	-58.98

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	148.35	kN
Mux	=	5.29	kNm
Muy	=	-58.98	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	84.88	deg
MRes	=	59.21	kNm
MCap	=	65.24	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.91 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	19
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	39.5525	-38.0427
Pu (kN)	159.46	148.35
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.4299	0.4697
Pt (%)	0.801	0.801
Design shear strength,	0.5897	0.5897

Tc (N/sqmm)		
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.208	1.2198
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.208	1.2198
Enhanced shear strength, Tc-e (N/sqmm)	0.7124	0.7193
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C12	
Level	:	5.029 m To 8.077 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm

Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	77	
Critical Analysis Load Combination	:	19	
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	103.41	kN
Mux	=	-6.24	kNm
Muy	=	51.6	kNm
Vux	=	-32.92	kN
Vuy	=	-3.54	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521
Top	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.7

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166
Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Sway Condition (as per Stability Index)	=	Non Sway	
Effective Length Factor along Minor axis	=	0.55	

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	2.09	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.7	
Slenderness Ratio	=	Effective Length / D	
	=	4.07, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.55	
Slenderness Ratio	=	Effective Length / B	
	=	6.26, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-6.24	2.09	-6.24	0	-6.24
Minor Axis - Muy	51.6	---	51.6	0	51.6

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	103.41	kN
Mux	=	-6.24	kNm
Muy	=	51.6	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	83.1	deg
MRes	=	51.98	kNm
MCap	=	63.3	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.82 <= 1	

Shear Calculation	Along D	Along B
Critical Analysis Load Combination	20	19
Critical Load Combination	[16] : 0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	33.1751	-32.9185
Pu (kN)	107.76	103.41
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3606	0.4064
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1406	1.1532
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength	1.5	1.5

Enhancement Factor (max)		
Shear Strength Enhancement Factor	1.1406	1.1532
Enhanced shear strength, Tc-e (N/sqmm)	0.6726	0.6801
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C12	
Level	:	8.077 m To 11.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	450	mm
Clear Cover	=	40	mm

Clear Floor Height @ B	=	2618	mm
Clear Floor Height @ D	=	2618	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.96 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	80	
Critical Analysis Load Combination	:	19	
Load Combination	=	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Critical Location	=	Top Joint	
Pu	=	49.26	kN
Mux	=	16.24	kNm
Muy	=	39.75	kNm
Vux	=	-23.34	kN
Vuy	=	6.44	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.521
Top	57.302	4496 x 230 x 430	2134 x 230 x 430	33.894	71.41	0.352

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.66

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.166

Top	14.969	1372 x 230 x 430	3925 x 230 x 430	111.07	38.825	0.091

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.54

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	20.24	mm
Minimum Eccentricity	>	20	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	1	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.38	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.66
Slenderness Ratio	=	Effective Length / D
	=	3.84, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.54
Slenderness Ratio	=	Effective Length / B
	=	6.15, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	16.24	1	16.24	0	16.24
Minor Axis - Muy	39.75	---	39.75	0	39.75

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	49.26	kN
Mux	=	16.24	kNm
Muy	=	39.75	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.6	
Reinforcement Provided	=	4-T20 + 2-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	67.77	deg
MRes	=	42.94	kNm
MCap	=	67.05	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.64 <= 1	

Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	15	19
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[15] : 0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	35.0964	-23.3352
Pu (kN)	92.16	49.26
Deff , Beff (mm)	400	180
Aeff (sqmm)	92000	81000
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.3815	0.2881
Pt (%)	0.801	0.801
Design shear strength, Tc (N/sqmm)	0.5897	0.5897
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.1202	1.073
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.1202	1.073
Enhanced shear strength, Tc-e (N/sqmm)	0.6606	0.6327
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor

Shear Links Design	Shear Links Not Required	Shear Links Not Required

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	5	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	Shear Design	Ductile Design	Normal Zone	Ductile Zone
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C13	
Level	:	11.125 m To 13.625 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	300	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2200	mm
Clear Floor Height @ D	=	2200	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	

D/B Ratio	:	1.3 <= 4 Hence, Design as Column

Flexural Design (Analysis Forces)

Analysis Reference No.	=	175	
Critical Analysis Load Combination	:	5	
Load Combination	=	[1] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Critical Location	=	Bottom Joint	
Pu	=	9.32	kN
Mux	=	-16.91	kNm
Muy	=	-7.4	kNm
Vux	=	-5.88	kN
Vuy	=	10.45	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	20.7	No Beam	No Beam	-	-	1
Top	20.7	2134 x 230 x 300	No Beam	24.25	-	0.461

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.81

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	12.167	No Beam	No Beam	-	-	1
Top	12.167	3924 x 230 x 300	No Beam	13.188	-	0.48

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.82

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	14.4	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.19	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	9.57	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.81
Slenderness Ratio	=	Effective Length / D
	=	5.94, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.82
Slenderness Ratio	=	Effective Length / B
	=	7.84, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-16.91	0.19	-16.91	0	-16.91
Minor Axis - Muy	-7.4	---	-7.4	0	-7.4

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	9.32	kN
Mux	=	-16.91	kNm
Muy	=	-7.4	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17

Reinforcement Provided	=	4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	23.63	deg
MRes	=	18.46	kNm
MCap	=	41.6	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.44 <= 1	

Shear Calculation (Analysis Forces)	**Along D**	**Along B**
Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	5	14
Critical Load Combination	[1] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	10.4521	-6.8844
Pu (kN)	9.32	8.88
Deff , Beff (mm)	252	182
Aeff (sqmm)	57960	54600
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.1803	0.1261
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0193	1.0195
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0193	1.0195
Enhanced shear strength, Tc-e (N/sqmm)	0.5302	0.5303
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	

	=	4	mm
Criterion for spacing of normal links			
Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C14	
Level	:	11.125 m To 13.625 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	300	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2200	mm
Clear Floor Height @ D	=	2200	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	176
Critical Analysis Load Combination	:	5
Load Combination	=	[1] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)

Critical Location	=	Bottom Joint	
Pu	=	0.44	kN
Mux	=	-24.98	kNm
Muy	=	4.27	kNm
Vux	=	3.97	kN
Vuy	=	16.26	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	20.7	No Beam	No Beam	-	-	1
Top	20.7	2134 x 230 x 300	No Beam	24.25	-	0.461

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.81

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	12.167	No Beam	No Beam	-	-	1
Top	12.167	No Beam	3924 x 230 x 300	-	13.188	0.48

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.82

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	14.4	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.01	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	9.57

	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.81	
Slenderness Ratio	=	Effective Length / D	
	=	5.94, Column not Slender Along D	

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.82	
Slenderness Ratio	=	Effective Length / B	
	=	7.84, Column Not Slender Along B	

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-24.98	0.01	-24.98	0	-24.98
Minor Axis - Muy	4.27	---	4.27	0	4.27

Where

A = Moments directly from analysis

B = Moments due to minimum eccentricity

C = Maximum of analysis moment and min. eccentricity = Max (A,B)

E = Moment due to slenderness effect

F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	0.44	kN
Mux	=	-24.98	kNm
Muy	=	4.27	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	9.7	deg
MRes	=	25.35	kNm
MCap	=	41.94	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.6 <= 1	

Shear Calculation (Analysis Forces)	**Along D**	**Along B**
Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	5	12
Critical Load Combination	[1] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	[8] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	16.2609	6.2805
Pu (kN)	0.44	3.49
Deff , Beff (mm)	252	182
Aeff (sqmm)	57960	54600
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.2806	0.115
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0009	1.0077
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0009	1.0077
Enhanced shear strength, Tc-e (N/sqmm)	0.5207	0.5242
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C15	
Level	:	13.625 m To 14.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	300	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2700	mm
Clear Floor Height @ D	=	2700	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	183	
Critical Analysis Load Combination	:	15	
Load Combination	=	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Critical Location	=	Bottom Joint	
Pu	=	12.32	kN
Mux	=	-19.77	kNm
Muy	=	-5.82	kNm
Vux	=	-3.09	kN
Vuy	=	11.02	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	103.5	No Beam	No Beam	-	-	1
Top	103.5	No Beam	4991 x 230 x 300	-	10.369	0.909

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Major Axis = 0.96

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	60.835	No Beam	No Beam	-	-	1
Top	60.835	No Beam	1372 x 230 x 300	-	37.719	0.617

Sway Condition (as per Stability Index) = Non Sway

Effective Length Factor along Minor axis = 0.86

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity = Unsupported Length / 500 + D / 30

= 15.4 mm

Mminx = Pu x Minimum Eccentricity

= 0.25 kNm

Slenderness Check

Max Slenderness Ratio(L/B) = 11.74

< 60 (Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor = 0.96

Slenderness Ratio = Effective Length / D

= 8.64, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.86
Slenderness Ratio	=	Effective Length / B
	=	10.1, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	-19.77	0.25	-19.77	0	-19.77
Minor Axis - Muy	-5.82	---	-5.82	0	-5.82

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	12.32	kN
Mux	=	-19.77	kNm
Muy	=	-5.82	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	4-T16	
Load Angle	=	Tan^{-1}(Muy/Mux)	
	=	16.4	deg
MRes	=	20.61	kNm
MCap	=	42.74	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.48 <= 1	

Shear Calculation (Analysis Forces)	Along D	Along B
Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	15	14
Critical Load Combination	[11] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	11.0157	-5.213

Pu (kN)	12.32	3.86
Deff , Beff (mm)	252	182
Aeff (sqmm)	57960	54600
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.1901	0.0955
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	
	1.0255	1.0085
Shear Strength Enhancement Factor (Tensile Force)	-	-
Shear Strength Enhancement Factor (max)	1.5	1.5
Shear Strength Enhancement Factor	1.0255	1.0085
Enhanced shear strength, Tc-e (N/sqmm)	0.5335	0.5246
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---

General Data

Column No.	:	C16	
Level	:	13.625 m To 14.125 m	
Frame Type	=	Non-Ductile	
Response Reduction Factor	=	3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	N/sqmm
Grade Of Steel (Main)	=	Fe550	N/sqmm
Grade Of Steel (Shear)	=	Fe550	N/sqmm
Grade Of Steel - Flexural Design	=	Fe550	N/sqmm
Grade Of Steel - Shear Design	=	Fe415	N/sqmm
Column B	=	230	mm
Column D	=	300	mm
Clear Cover	=	40	mm
Clear Floor Height @ B	=	2700	mm
Clear Floor Height @ D	=	2700	mm
No Of Floors	=	1	
No Of Columns In Group	=	1	
Column Type	:	UnBraced	
Minimum eccentricity check	:	One Axis at a Time	
Code defined D/B ratio	:	4	
D/B Ratio	:	1.3 <= 4 Hence, Design as Column	

Flexural Design (Analysis Forces)

Analysis Reference No.	=	184	
Critical Analysis Load Combination	:	13	
Load Combination	=	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Critical Location	=	Bottom Joint	
Pu	=	8.51	kN
Mux	=	18.62	kNm
Muy	=	-7.03	kNm
Vux	=	-3.87	kN
Vuy	=	-10.24	kN

Effective Length Calculation

Calculation Along Major Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	103.5	No Beam	No Beam	-	-	1
Top	103.5	4991 x 230 x 300	No Beam	10.369	-	0.909

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Major Axis	=	0.96

Calculation Along Minor Axis Of Column

Joint	Column Stiffness	Beam Sizes		Beam Stiffness		Beta
		Beam 1 (Length x Width x Depth)	Beam 2 (Length x Width x Depth)	Beam 1	Beam 2	
	N-m x 10^6	mm	mm	N-m x 10^6	N-m x 10^6	
Bottom	60.835	No Beam	No Beam	-	-	1
Top	60.835	No Beam	1372 x 230 x 300	-	37.719	0.617

Sway Condition (as per Stability Index)	=	Non Sway
Effective Length Factor along Minor axis	=	0.86

Minimum Eccentricity Check

Since Axial Force is compressive, Min. Eccentricity check to be performed

Most critical case is with Min. Eccentricity check in X-direction

Minimum Eccentricity Along D:

Minimum Eccentricity	=	Unsupported Length / 500 + D / 30	
	=	15.4	mm
Mminx	=	Pu x Minimum Eccentricity	
	=	0.17	kNm

Slenderness Check

Max Slenderness Ratio(L/B)	=	11.74	
	<	60	(Hence Ok)

Column Is Unbraced Along D

Slenderness Check Along D:

Effective Length Factor	=	0.96
Slenderness Ratio	=	Effective Length / D
	=	8.64, Column not Slender Along D

Column Is Unbraced Along B

Slenderness Check Along B:

Effective Length Factor	=	0.86
Slenderness Ratio	=	Effective Length / B
	=	10.1, Column Not Slender Along B

Calculation of Design Moment

Direction	Manalysis	Mmin (Abs)	Mdesign	Mslndx (Abs)	Mdesign-final
	A	B	C	E	F
Major Axis - Mux	18.62	0.17	18.62	0	18.62

Minor Axis - Muy	-7.03	---	-7.03	0	-7.03

Where

A = Moments directly from analysis
B = Moments due to minimum eccentricity
C = Maximum of analysis moment and min. eccentricity = Max (A,B)
E = Moment due to slenderness effect
F = Final design Moment = Max(C- Top Bottom , D- Top Bottom) + E

Final Critical Design Forces

Pu	=	8.51	kN
Mux	=	18.62	kNm
Muy	=	-7.03	kNm

Resultant Moment (Combined Action)

Moment Capacity Check

Pt Calculated	=	1.17	
Reinforcement Provided	=	4-T16	
Load Angle	=	$Tan^{-1}(Muy/Mux)$	
	=	20.69	deg
MRes	=	19.9	kNm
MCap	=	41.94	kNm
Capacity Ratio	=	MRes/ MCap	
	=	0.47 <= 1	

Shear Calculation (Analysis Forces)	**Along D**	**Along B**
Shear Calculation	**Along D**	**Along B**
Critical Analysis Load Combination	13	14
Critical Load Combination	[9] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	[10] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
Design shear force, Vu (kN)	-10.2427	-6.1878
Pu (kN)	8.51	-0.3
Deff , Beff (mm)	252	182
Aeff (sqmm)	57960	54600
Design shear stress, Tv	Vu / (B x Deff)	Vu / (D x Beff)
Tv (N/sqmm)	0.1767	0.1133
Pt (%)	0.583	0.583
Design shear strength, Tc (N/sqmm)	0.5202	0.5202
Shear Strength Enhancement Factor (Compressive Force)	1 + 3 x Pu / (B x D x Fck)	

	1.0176	-
Shear Strength Enhancement Factor (Tensile Force)	-	1
Shear Strength Enhancement Factor (max)	1.5	-
Shear Strength Enhancement Factor	1.0176	-
Enhanced shear strength, Tc-e (N/sqmm)	0.5294	0.5202
Design shear check	Tv < Tc x Enhancement factor	Tv < Tc x Enhancement factor
Shear Links Design	**Shear Links Not Required**	**Shear Links Not Required**

Design Of Links

Links in the zone where special confining links are not required

Normal Links

Diameter of link	=	8	mm
	>	Max. longitudinal bar dia / 4	
	=	4	mm

Criterion for spacing of normal links

Min. Longitudinal Bar dia X 16	=	256	mm
Min. dimension of column	=	230	mm
Max. 300 mm	=	300	mm
Provided spacing	=	225	mm

Table For Links

	Required			Provided	
	Normal Design	**Shear Design**	**Ductile Design**	**Normal Zone**	**Ductile Zone**
Link Dia.	8	---	---	8	---
Spacing	225	---	---	225	---